The Physical Layer

Chapter 2

2.2 - 2.4

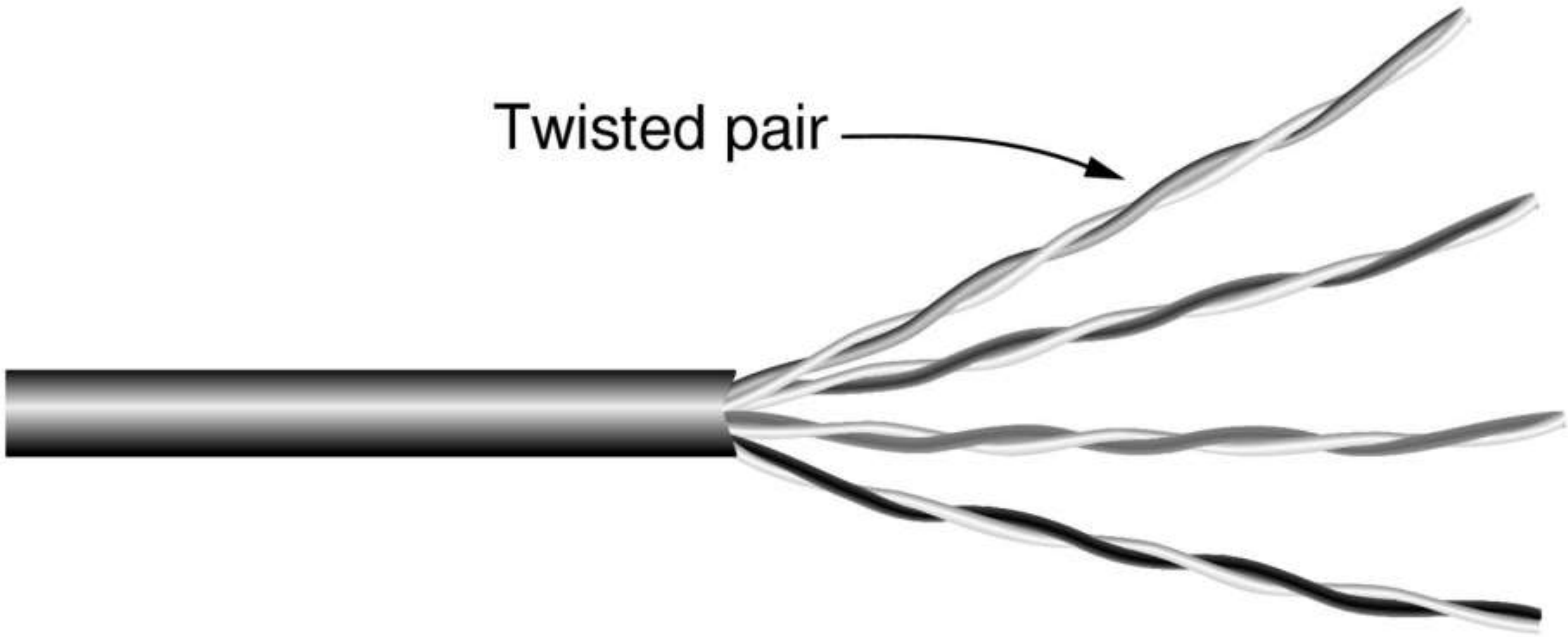
Guided Transmission Media

- Magnetic media
- Twisted pairs
- Coaxial cable
- Power lines
- Fiber optics

Magnetic Media

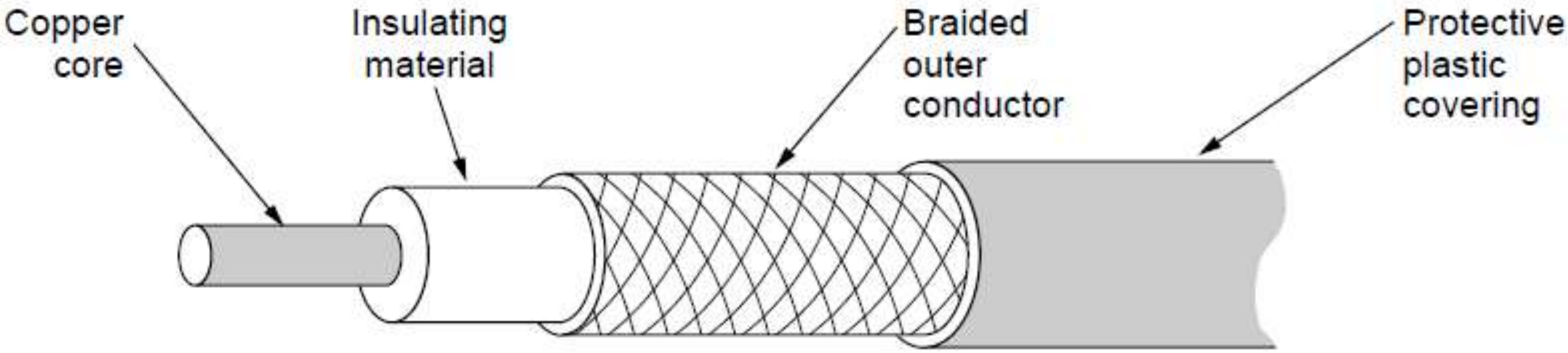
- Write data onto magnetic media
  - Disks
  - Tapes
- Data transmission speed
  - Never underestimate the bandwidth of a station wagon full of tapes hurtling down the highway.

Twisted Pairs



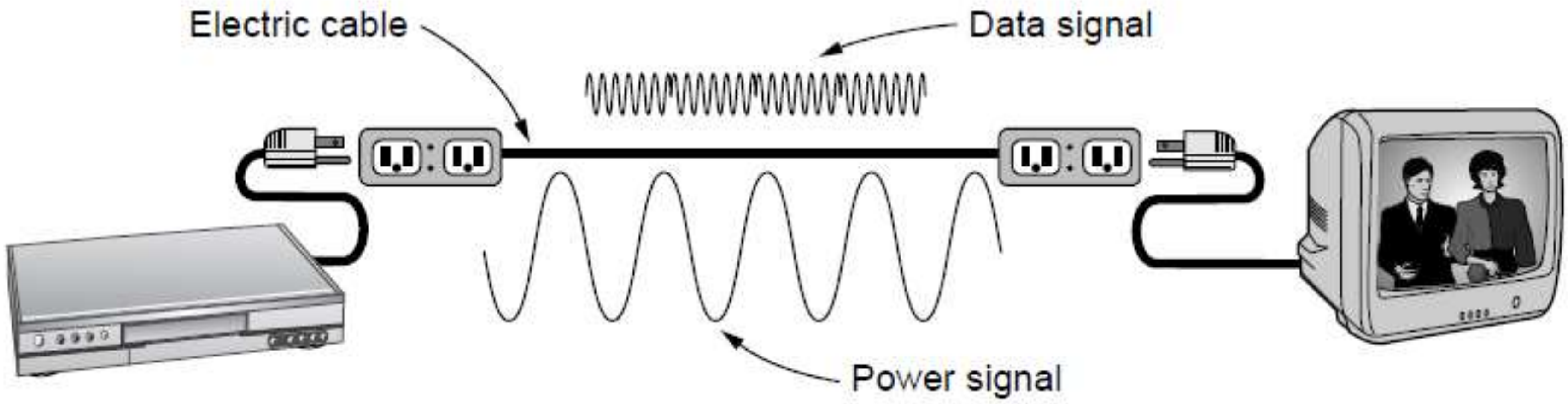
Category 5 UTP cable with four twisted pairs

Coaxial Cable



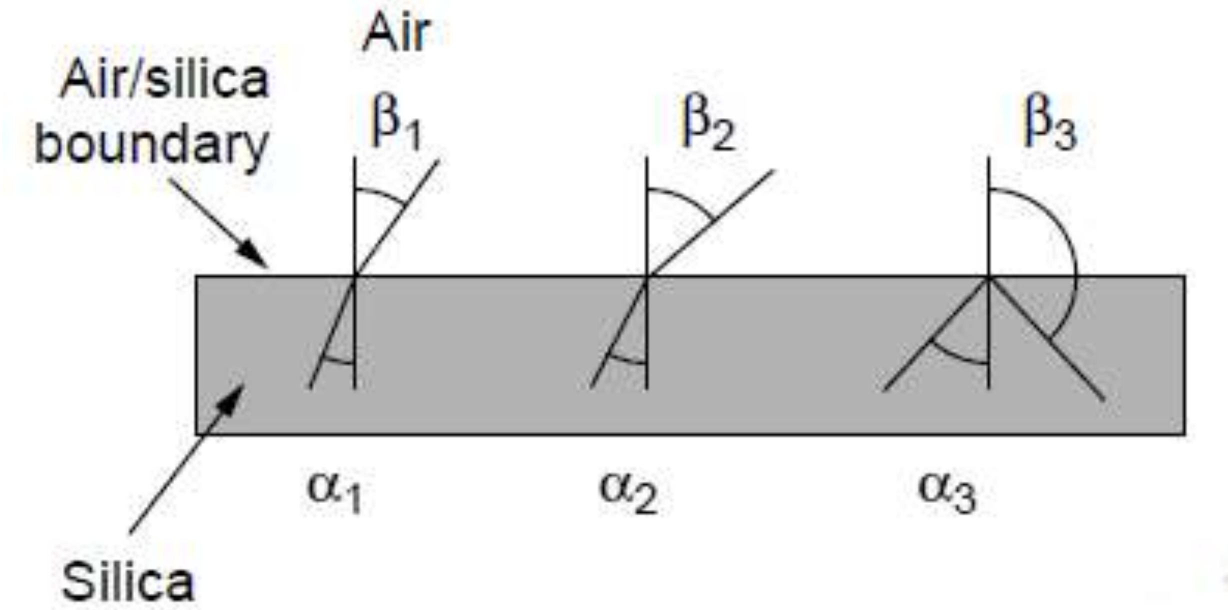
A coaxial cable

Power Lines



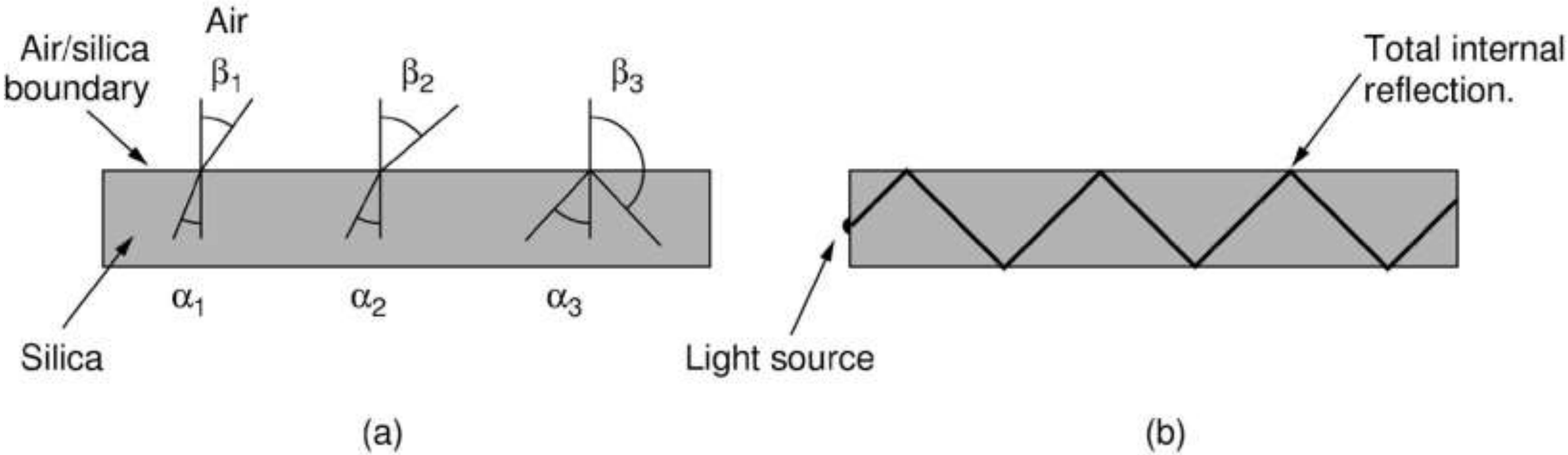
A network that uses household electrical wiring.

Fiber Optics (1)



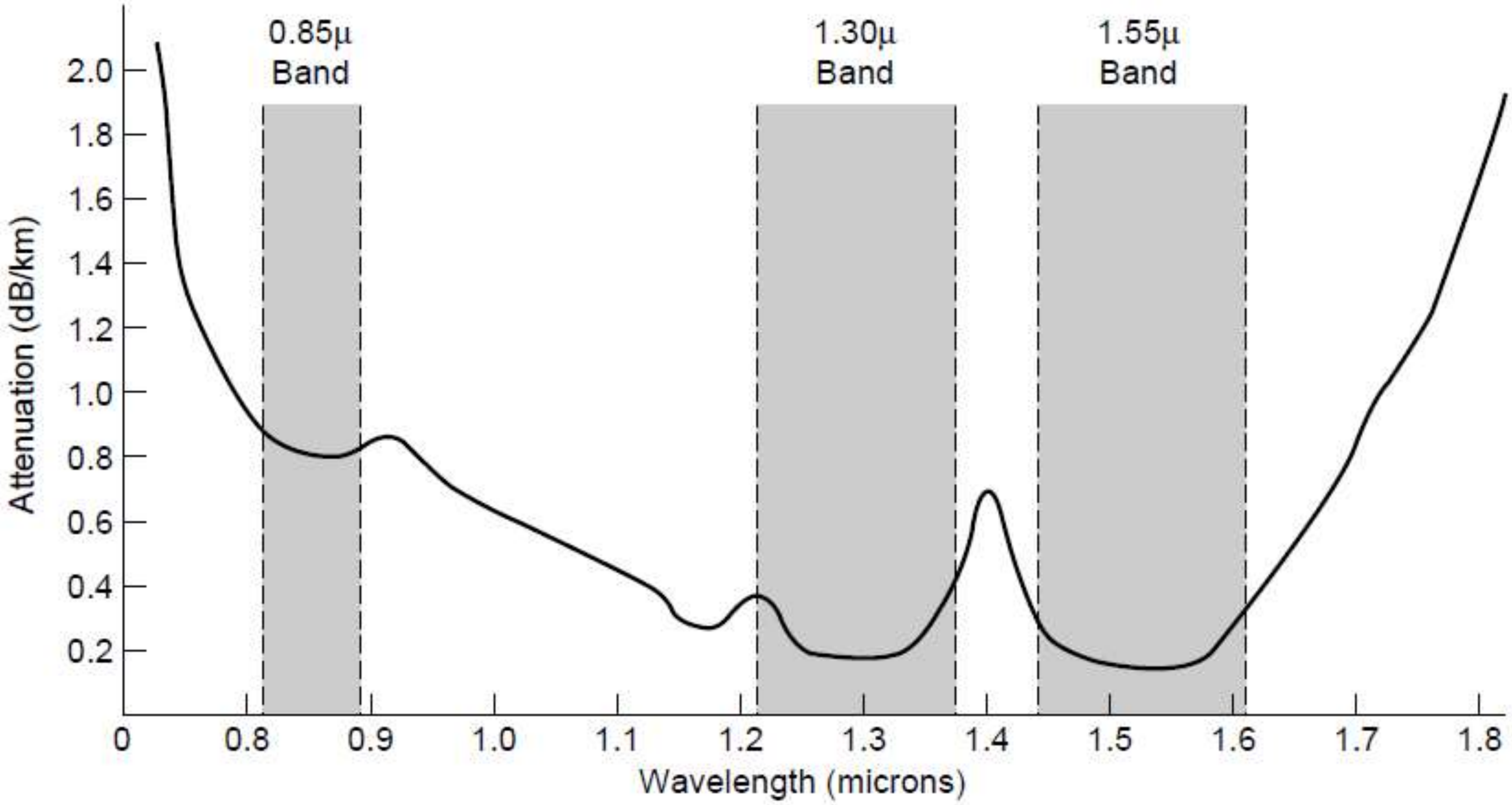
Three examples of a light ray from inside a silica fiber impinging on the air/silica boundary at different angles.

Fiber Optics (2)



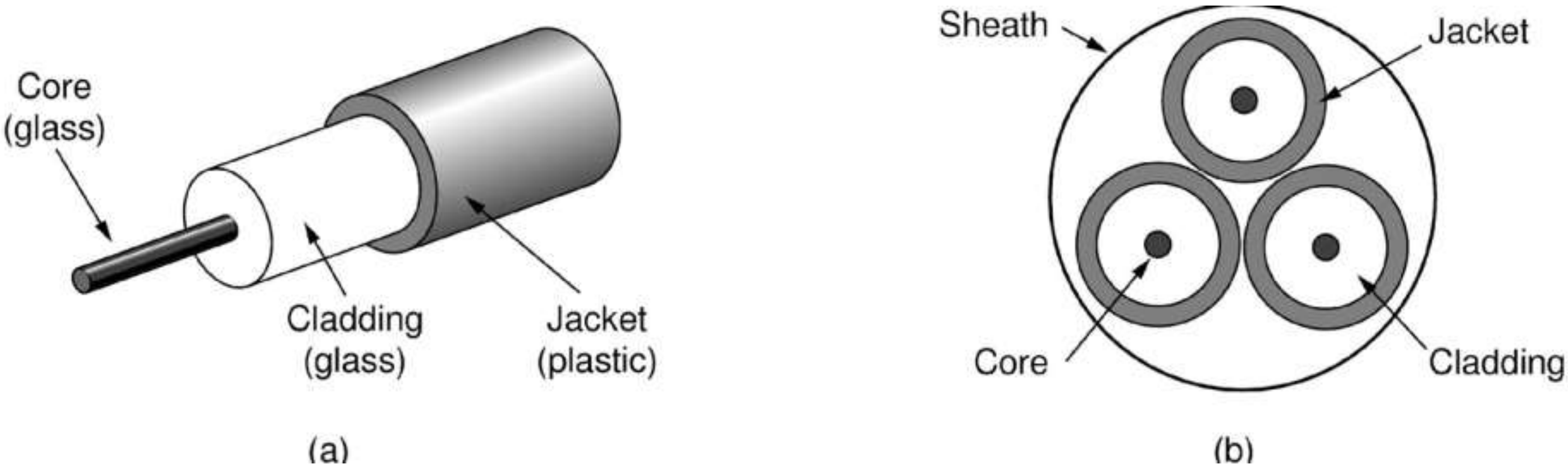
Light trapped by total internal reflection.

Transmission of Light Through Fiber



Attenuation of light through fiber in the infrared region

Fiber Cables (1)



Views of a fiber cable

Fiber Cables (2)

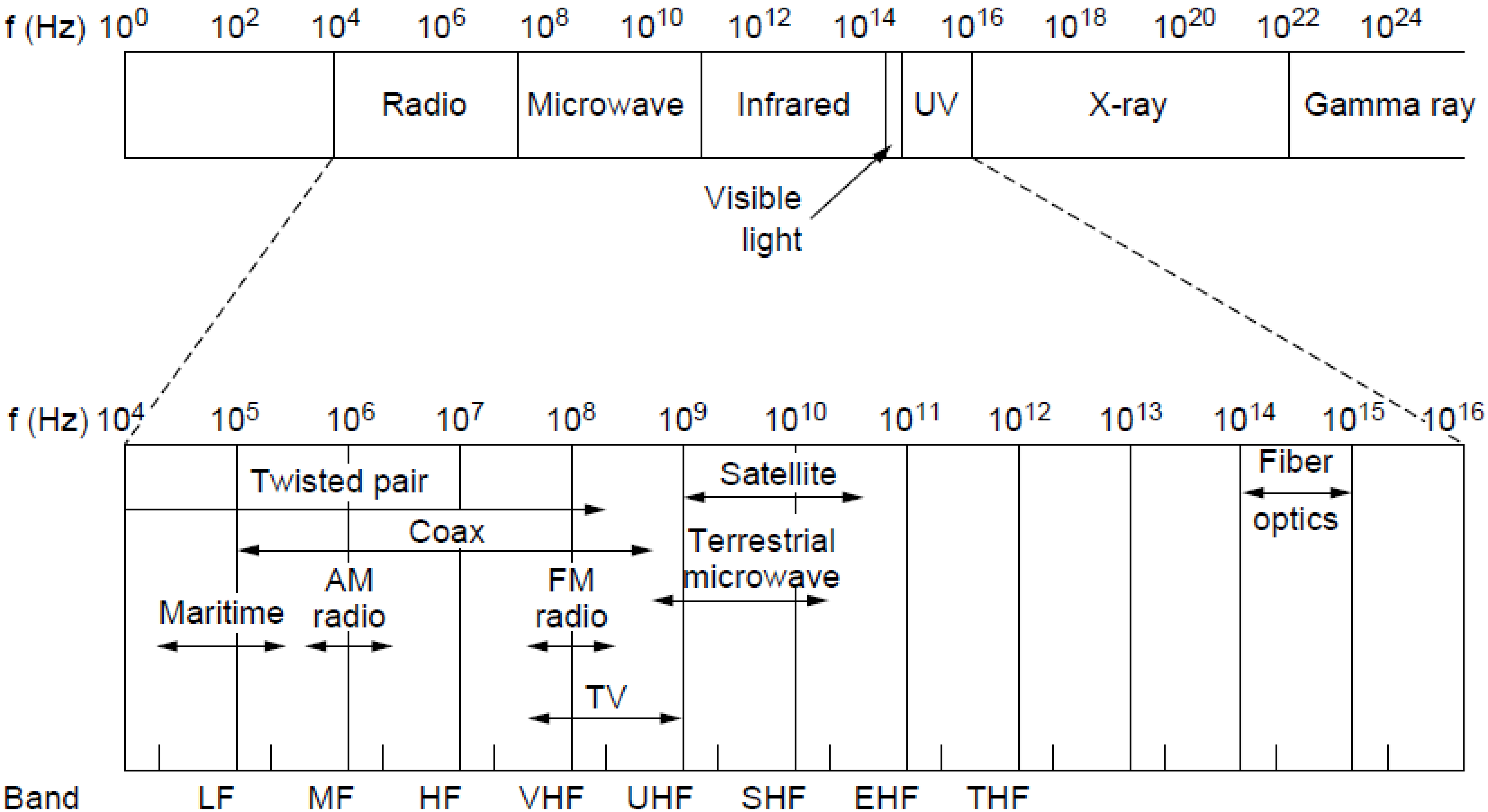
Item	LED	Semiconductor laser
Data rate	Low	High
Fiber type	Multi-mode	Multi-mode or single-mode
Distance	Short	Long
Lifetime	Long life	Short life
Temperature sensitivity	Minor	Substantial
Cost	Low cost	Expensive

A comparison of semiconductor diodes and LEDs as light sources

Wireless Transmission

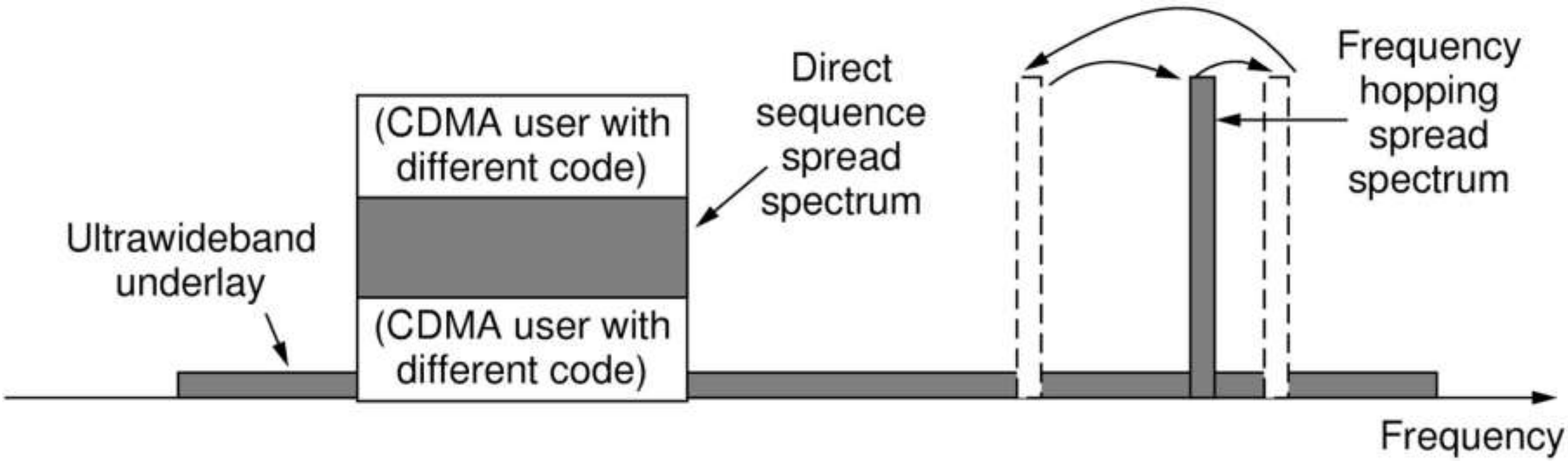
- The Electromagnetic Spectrum
- Radio Transmission
- Microwave Transmission
- Infrared Transmission
- Light Transmission

The Electromagnetic Spectrum (1)



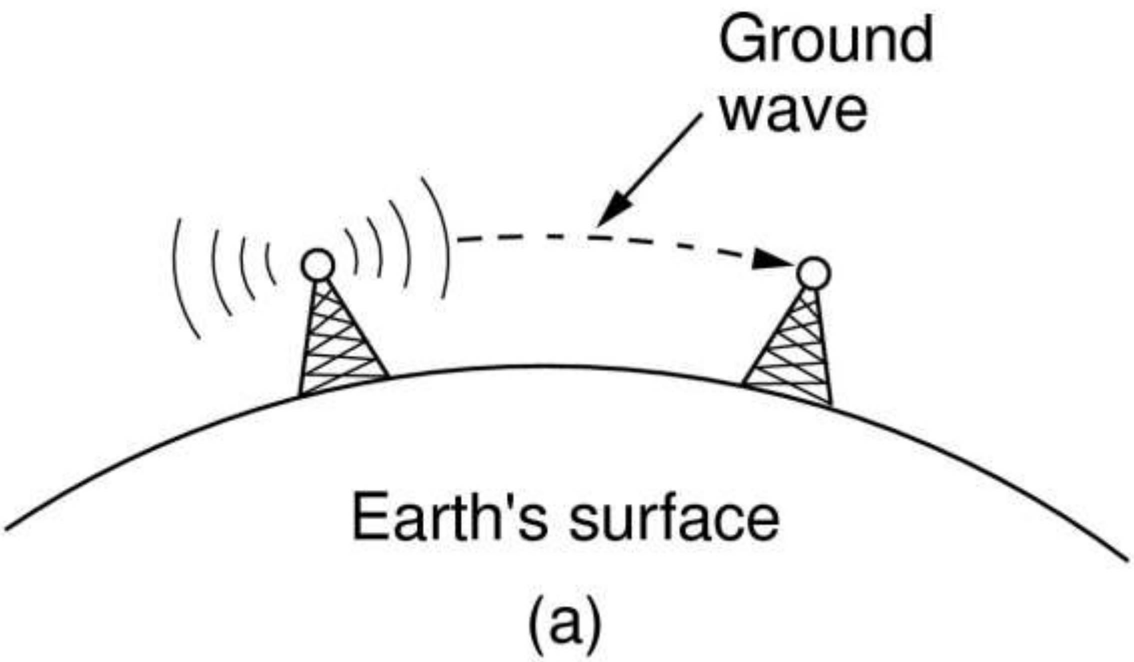
The electromagnetic spectrum and its uses for communication

The Electromagnetic Spectrum (2)



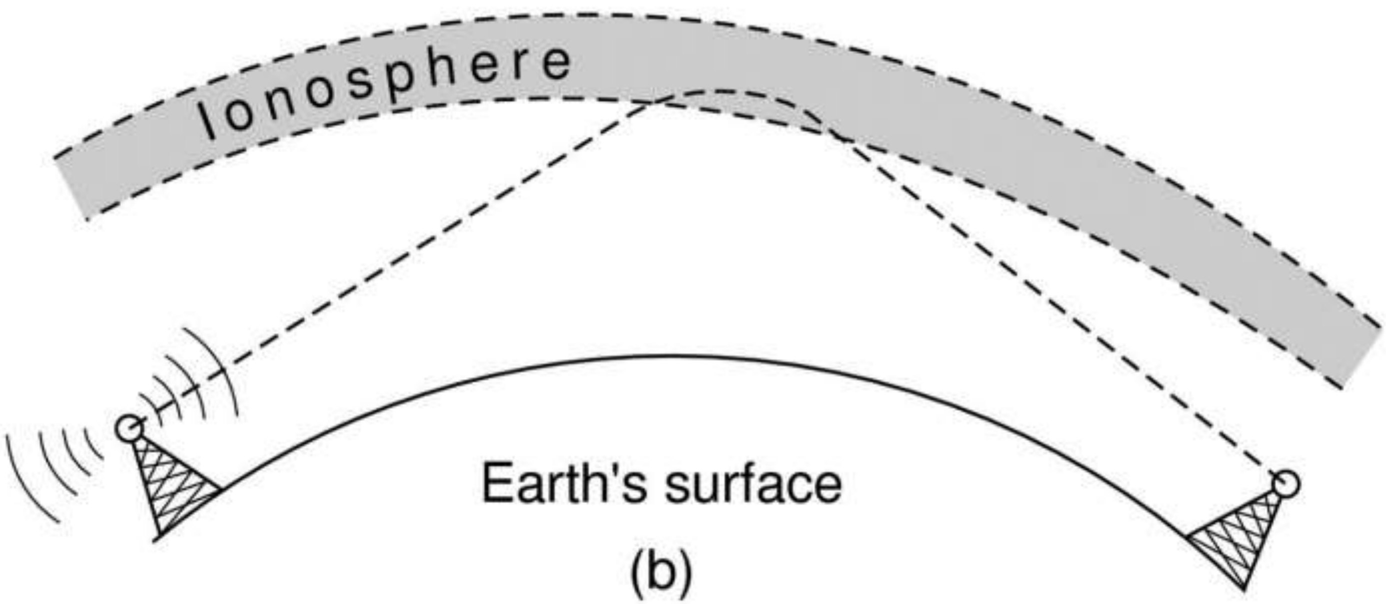
Spread spectrum and ultra-wideband (UWB) communication

Radio Transmission (1)



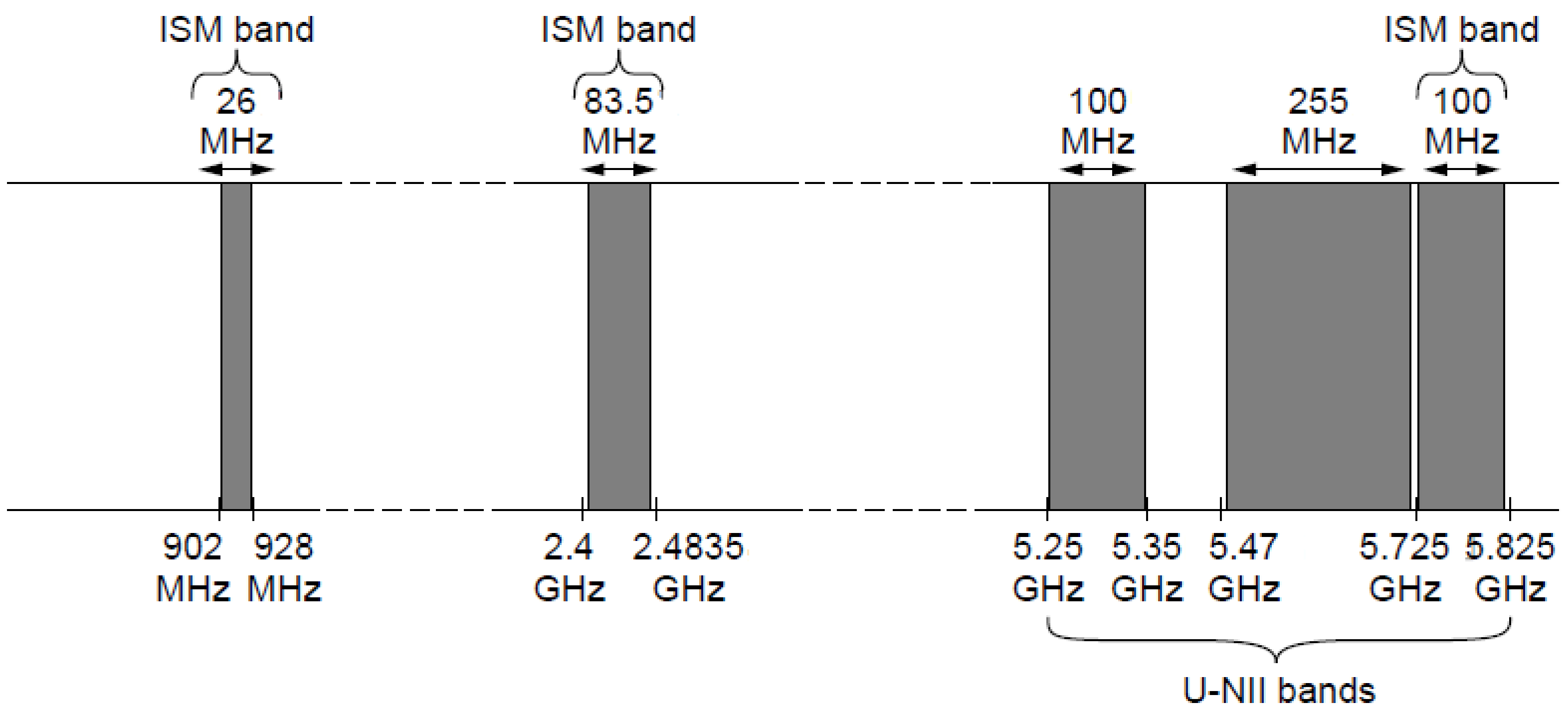
In the VLF, LF, and MF bands, radio waves follow the curvature of the earth

Radio Transmission (2)



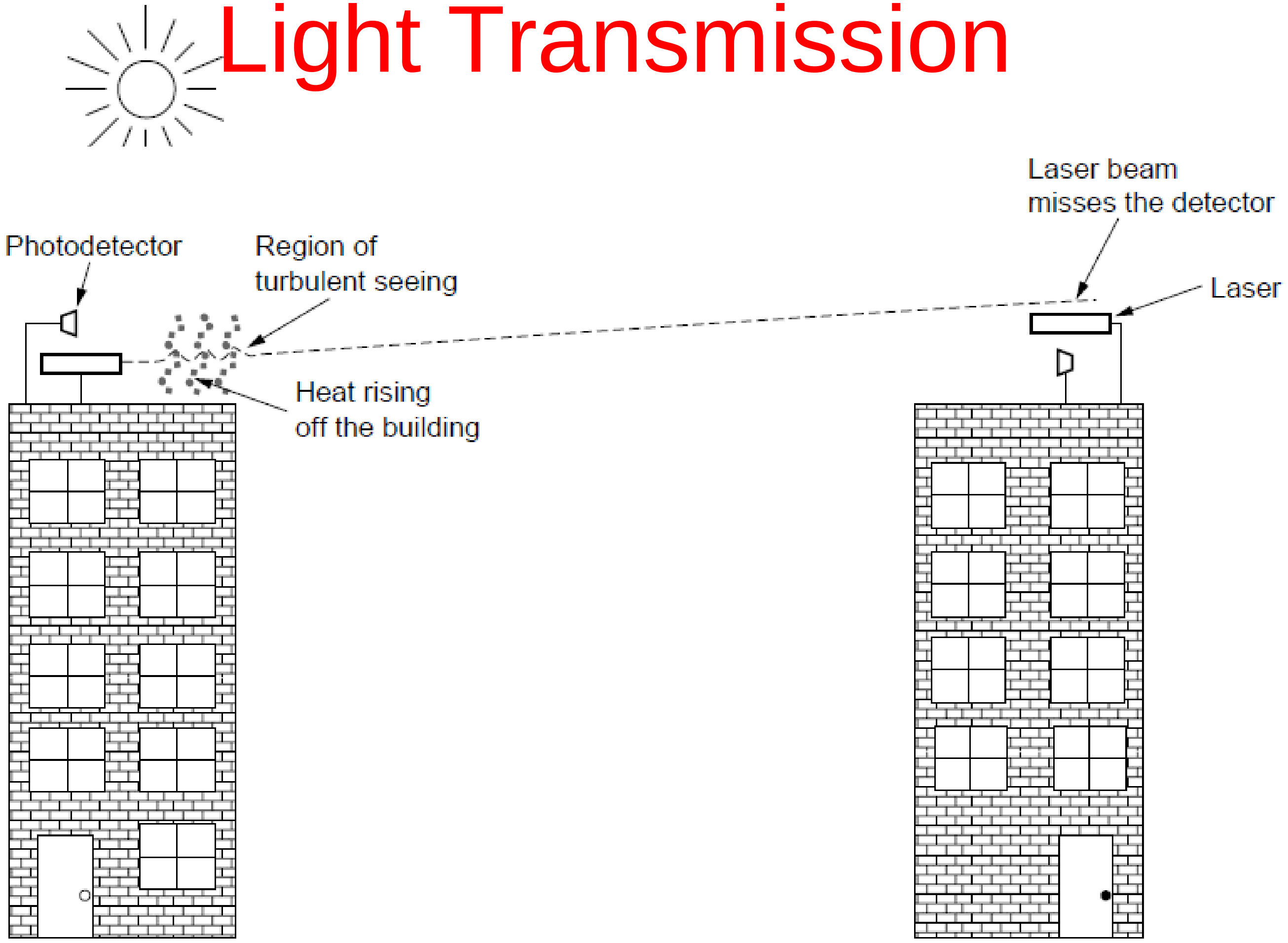
In the HF band, they bounce off the ionosphere.

The Politics of the Electromagnetic Spectrum



ISM and U-NII bands used in the United States by wireless devices

Light Transmission



Convection currents can interfere with laser communication systems. A bidirectional system with two lasers is pictured here.